

dot Product

- the dot Product is scalar it has magnitude but does Not have direction
or of two vectors term Vector $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{B} = \langle b_1, b_2, b_3 \rangle$

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

• you just add these together you do Not need to include $\hat{i}$, $\hat{j}$ and $\hat{k}$

- Properties of dot Product

a) $\vec{a} \cdot \vec{a} = |\mathbf{a}|^2$

b) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

c) $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$

d) $(c\vec{a}) \cdot \vec{b} = c \cdot (\vec{a} \cdot \vec{b}) = \vec{a} \cdot (c\vec{b})$

- a Dot Product is a scalar quantity made of the product of the magnitudes of two vectors and cosine of the angle between



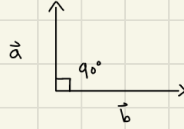
you can also write the dot Product like:

$$\vec{A} \cdot \vec{B} = |\vec{A}| \cdot |\vec{B}| \cos(\theta)$$

you can also use this equation to Find the angle by Just isolating Cosine

$$\theta = \cos^{-1} \left(\frac{\vec{A} \cdot \vec{B}}{|\vec{A}| \cdot |\vec{B}|} \right)$$

- two vectors $\vec{a}$ and $\vec{b}$ are called perpendicular or orthogonal if the angle between them is 90° or $\pi/2$



- Orthogonal Means they meet at a right angle or 90° degrees

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta \Rightarrow \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos(90^\circ)$$

$$\vec{a} \cdot \vec{b} = 0$$

- this only happens when you have perpendicular vectors

if you have parallel vectors in the same direction



$$\text{then } |\vec{a}| |\vec{b}| \cos(0) \Rightarrow \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}|$$

if you have parallel and points in the opposite direction



$$\text{then } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = |\vec{a}| |\vec{b}| \cos(180^\circ) = \vec{a} \cdot \vec{b} = -|\vec{a}| |\vec{b}|$$

examples:

Find the dot product of $\vec{a} = \langle 2, -3, 4 \rangle$ and $\vec{b} = \langle -1, 2, 5 \rangle$

$$\vec{a} \cdot \vec{b} = (2)(-1) + (-3)(2) + (4)(5) = \boxed{12}$$

* a dot product of two vectors is scalar

$$\vec{a} \cdot \vec{a} = (2)^2 + (-1)^2 + (4)^2 = \boxed{21}$$

The scalar projection of $\vec{b}$ on to $\vec{a}$

$$\text{Comp}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

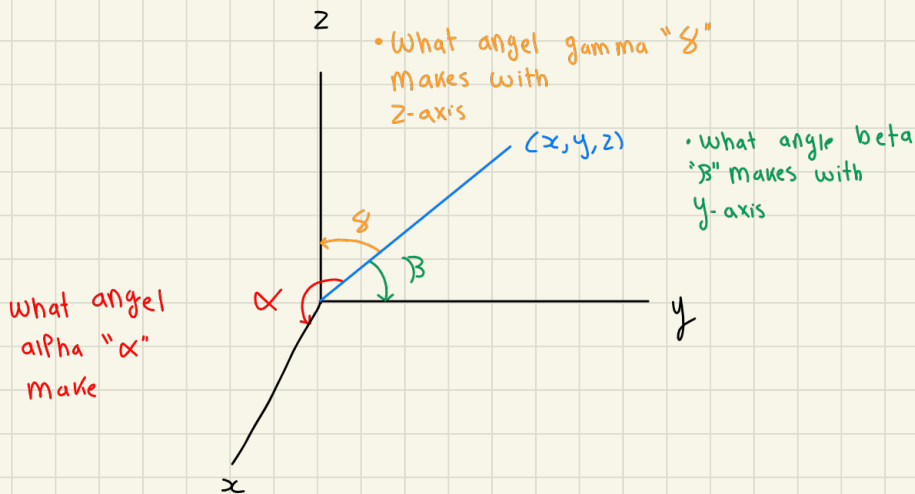
the vector projection of $\vec{b}$ on to $\vec{a}$

$$\text{Proj}_{\vec{a}} \vec{b} = \frac{(\vec{a} \cdot \vec{b}) \cdot \vec{a}}{|\vec{a}|^2} \quad \text{or} \quad \text{Proj}_{\vec{a}} \vec{b} = (\text{Comp}_{\vec{a}} \vec{b}) \cdot \frac{\vec{a}}{|\vec{a}|}$$

either one works

the orthogonal projection

$$\text{Orth}_{\vec{a}} \vec{b} = \vec{b} - \text{Proj}_{\vec{a}} \vec{b}$$



$$\vec{a} = \langle x, y, z \rangle = \langle |\vec{a}| \cos(\alpha), |\vec{a}| \cos(\beta), |\vec{a}| \cos(\gamma) \rangle$$

$$\frac{\vec{a}}{|\vec{a}|} = \langle \cos \alpha, \cos \beta, \cos \gamma \rangle$$

↑ unit vector

• You can then find the unit Vector then:

$$\frac{\vec{a}}{|\vec{a}|} = \cos \alpha$$

$$\frac{\vec{a}}{|\vec{a}|} = \cos \beta$$

$$\frac{\vec{a}}{|\vec{a}|} = \cos \gamma$$

• you can then isolate
by applying inverse cosine
For either α , β , or γ
and solve for it

$\vec{a} \cdot (\vec{b} \cdot \vec{c})$ dot product
↳ Vector

$\Rightarrow$ Vector $\cdot$ Scalar = Not Possible

$\vec{a} + (\vec{b} \cdot \vec{c})$ dot product
↑ Vector

$\Rightarrow$ cannot add or subtract a number
to a vector

the scalar projection of $\vec{b}$ on to $\vec{a}$

$$\text{comp}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$$

the vector projection of $\vec{b}$ on to $\vec{a}$

$$\text{proj}_{\vec{a}} \vec{b} = \frac{(\vec{a} \cdot \vec{b}) \cdot \vec{a}}{\|\vec{a}\|^2}$$

magnitude square
is the dot product $\vec{a} \cdot \vec{a}$

either one works

the orthogonal projection

$$\text{orth}_{\vec{a}} \vec{b} = \vec{b} - \text{proj}_{\vec{a}} \vec{b}$$

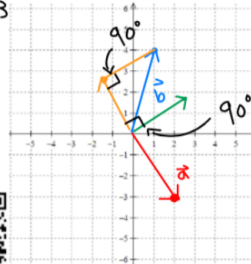
4. Find the scalar projection, vector projection and orthogonal projection of $\vec{b}$ onto $\vec{a}$

a. $\vec{a} = (2, -3)$ and $\vec{b} = (1, 4)$ (Sketch the vectors and its projections)

$$\text{comp}_{\vec{a}} \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} \quad |\vec{a}| = \sqrt{(2)^2 + (-3)^2} = \sqrt{13}$$

$$\vec{a} \cdot \vec{b} = (2)(1) + (-3)(4) = -10$$

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = \frac{-10}{\sqrt{13}} = -2.773$$



$$\text{proj}_{\vec{a}} \vec{b} = \frac{(\vec{a} \cdot \vec{b}) \vec{a}}{|\vec{a}|^2} \quad \text{or} \quad (\text{comp}_{\vec{a}} \vec{b}) \cdot \frac{\vec{a}}{|\vec{a}|}$$

$$\left(\frac{-10}{\sqrt{13}} \right) \cdot \left(\frac{2}{\sqrt{13}}, \frac{-3}{\sqrt{13}} \right) = \left\langle -\frac{20}{13}, \frac{30}{13} \right\rangle \approx \langle -1.5, 2.3 \rangle$$

$$\text{orth}_{\vec{a}} \vec{b} = \vec{b} - \text{Proj}_{\vec{a}} \vec{b}$$

$$\langle 1, 4 \rangle - \langle -1.5, 2.3 \rangle$$

$$\langle 1 + 4j \rangle, \langle 1.5i - 2.3j \rangle$$

$$\langle 2.5i + 1.7j \rangle = \langle 2.5, 1.7 \rangle$$

• application

$$\text{Work: } \vec{F} \cdot \vec{d} = |\vec{F}| |\vec{d}| \cos \theta$$

5. Find the work done by the force $\vec{F} = 8\vec{i} - 6\vec{j} + 9\vec{k}$ that moves an object from the point (0,5,4) along the straight line to the point (6,12,20). The distance is measured in meters and force in Newtons.

6. A tow truck drags a stalled car along a road. The chain makes an angle of 30° with the road and the tension on the chain is 1500N. How much work is done by the truck in pulling the car 1km.

5)

$$\begin{aligned} \vec{F} &= 8\vec{i} - 6\vec{j} + 9\vec{k} & \text{object moves from point } (0,5,4) & \text{ to } (6,12,20) \\ & & (6,12,20) - (0,5,4) & \\ & & \vec{d} = \langle 6, 7, 16 \rangle & \end{aligned}$$

$$\begin{aligned} \vec{a} \cdot \vec{b} &= a_1b_1 + a_2b_2 + a_3b_3 & (\text{we use this because } \theta & \text{ wasn't given}) \\ & (8)(6) + (-6)(7) + (9)(16) & \\ & = 158 \text{ Joules} & \end{aligned}$$

6)

$$\text{Work: } \vec{F} \cdot \vec{d} = |\vec{F}| |\vec{d}| \cos \theta$$



$$|\vec{F}| = 1500 \text{ N}$$

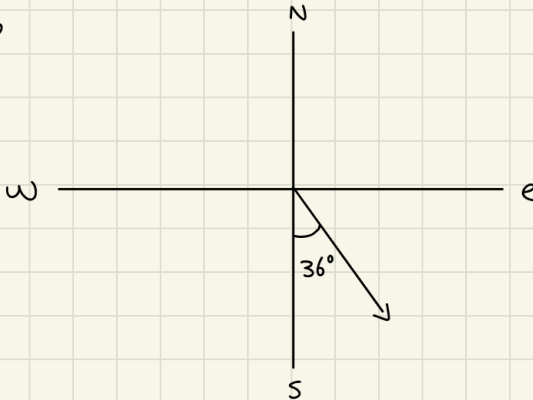
$$|\vec{d}| = 1 \text{ km (convert to meters)} = 1000 \text{ m}$$

$$\text{Work: } (1500) (1000) (\cos 30^\circ)$$

$$= 1299038.11 \text{ Joules}$$

8. A boat sails south with the help of a wind blowing in the direction $S36^\circ E$ with magnitude 400 lb . Find the work done by the wind as the boat moves 120 ft .

$$\text{Work: } |\vec{F}| |\vec{d}| \cos \theta$$



$$|\vec{F}| = 400 \text{ N}$$

$$|\vec{d}| = 120 \text{ ft}$$

$$(400) (120) (\cos 36^\circ)$$

$$= 38832.8 \text{ ft} \cdot \text{lb}$$